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COMET FWC 070

GATE QUESTION

Contents

1 Components 1
1.1 Breadboard . . . 1
1.2 Seven Segment Display . . . 1
1.3 Arduino . . . 2

2 Display Control through Hardware 2
2.1 Powering the Display . . . 2
2.2 Controlling the Display . . . 2

3 Display Control through Software 3

1 Components

Component	Value	Quantity
Breadboard	–	1
Resistor	$\geq 220\Omega$	1
Arduino Uno	–	1
Seven Segment Display	Common Anode	1
Jumper Wires	–	20

Table 1: Required Components

1.1 Breadboard

The breadboard can be divided into five segments. In each of the green segments, the pins are internally connected so as to have the same voltage.

Similarly, in the central segments, the pins in each column are internally connected in the same fashion as the blue columns.

1.2 Seven Segment Display

The seven segment display has eight pins:

$$a,\ b,\ c,\ d,\ e,\ f,\ g,\ \text{dot}$$

These pins take an active LOW input, i.e., the LED will glow only if the input is connected to ground. Each of these pins is connected to an LED segment. The dot pin is reserved for the decimal-point LED.

1.3 Arduino

The Arduino Uno has ground pins, analog input pins A0–A3, and digital pins D1–D13 that can be used for both input and output. It also has two power pins that can generate 3.3V and 5V.

In the following exercises, only the GND, 5V, and digital pins will be used.

2 Display Control through Hardware

2.1 Powering the Display

Problem 2.1

Plug the display into the breadboard and make the connections shown in Table 2.

Table 2: Arduino to Breadboard Connections

Arduino	Breadboard
5V	Top Green Rail
GND	Bottom Green Rail

Henceforth, all 5V and GND connections will be made from the breadboard.

Problem 2.2

Make the connections shown in Table 3.

Table 3: Breadboard to Display Connections

Breadboard	Display
5V	Resistor → COM
GND	DOT

Problem 2.3

Connect the Arduino to the computer. The DOT LED should glow.

2.2 Controlling the Display

The seven-segment display can be used to generate decimal digits. **Problem 2.4**

Generate the number 1 on the display by connecting only the pins b and c to GND (=0).

This corresponds to the first row of Table 4. A value of 1 means not connecting to GND.

Problem 2.5

Repeat the above exercise to generate the number 2 on the display.

Problem 2.6

Draw the numbers 0–9 using the seven segment display and complete Table 4.

Table 4: Digit Representation

a	b	c	d	e	f	g	Decimal
1	0	0	1	1	1	1	1
0	0	1	0	0	1	0	2

3 Display Control through Software

Problem 3.1

Make the connections according to Table 5.

Table 5: Arduino to Display Connections

Arduino	2	3	4	5	6	7	8
Display	a	b	c	d	e	f	g

Problem 3.2

Download and execute the following code using the Arduino IDE:

Problem 3.3

Generate the numbers 0–9 by modifying the above program.